

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #5

Instructions

- TA in-charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

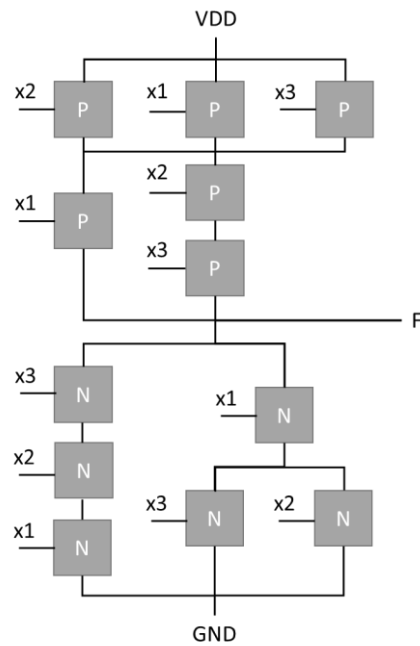
Question 1 (20%)	Question 2 (20%)	Question 3 (20%)	Total
Question 4 (20%)	Question 5 (20%)	-	
		-	

Question 1 (20%)

Design an SR-latch using only one D-latch and other logic gates. Note that both SR- and D-latches are not gated (no clk input).

Question 2 (20%)

For the circuit below, provide the corresponding truth table and derive a minimized POS expression. Implement the minimized expression using P and N switches. Note, each box in the figure represents a single P/N switch.



Question 3 (20%)

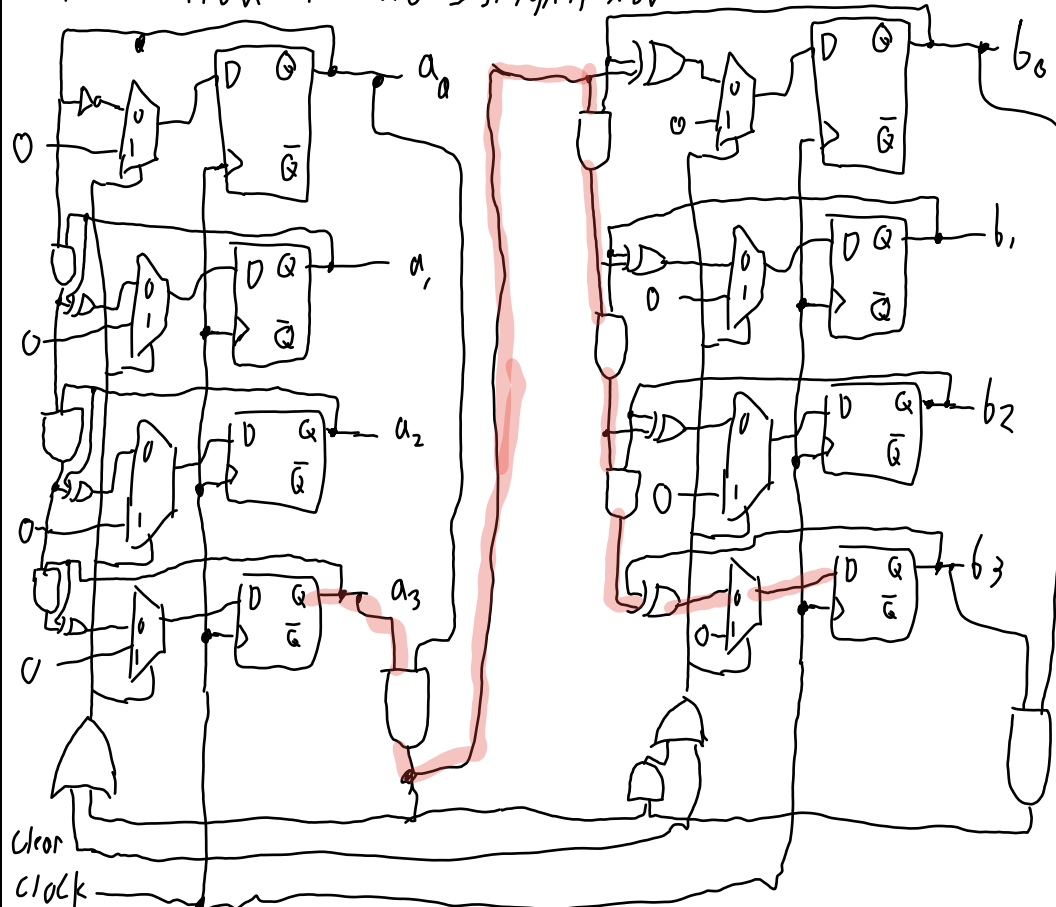
Determine the maximum operating frequency of a two-digit BCD up counter (as shown in class). Provide the circuit diagram of the counter and mark the critical path of the circuit.

Given the following specifications:

- FFs setup time is 3 ns
- FFs hold time is 1 ns
- FFs t_{CQ} and propagation delay of all gates and logic blocks is 3 ns
- The maximum fan-in of all gates is 2

Answer

Circuit from textbook - SI, slightly modified (removed first enable) note shown in class



Critical path, in red

$$T \geq t_{CQ} + t + t_{su}$$

$$T \geq 3ns + (3ns \cdot 6 \text{ gates/blocks}) + 3ns = 24ns$$

$$\text{Frequency max is } 41.6 \text{ MHz} = \frac{1}{24ns}$$

Question 4 (20%)

The 4-bit ring counter shown in class (implemented using DFFs) is operated using a clock frequency of 500 MHz, check the circuit for timing violations (show your work). If violations exist, correct the circuit using as few logic gates as possible (provide new schematic). Prove that all timing violations are solved and find the new operating frequency.

Given the following timing constraints:

- FFs setup time is 4 ns
- FFs hold time is 3 ns
- FFs t_{CQ} is 1 ns
- Latency of any gate is equal to the fan-in of the gate in units of ns

Answer

Original schematic from textbook

At 500 MHz
 $T = 2 \text{ ns}$

Setup time violation here
 $T \geq t_{CQ} + t_i + t_{su}$ Violated setup time condition
 $2 \text{ ns} \geq 1 \text{ ns} + 0 + 4 \text{ ns}$ $T_{old} \geq 5 \text{ ns}$ $T_{new} \geq 1 \text{ ns} + 2 \text{ ns} + 4 \text{ ns}$
 $T_{new} \geq 7 \text{ ns}$

hold time check
 $t_{CQ} + t_i \geq t_h$
 $1 \text{ ns} + 0 \geq 3 \text{ ns}$ also violated
 need $\geq 2 \text{ ns}$ of inverted Q + delay on all lines highlighted

$F_{max} \leq 142.8 \text{ MHz} = 1/7 \text{ ns}$

Question 5 (20%)

Design a 4-bit counter that counts in Excess-3 code. The counter should count to $(6)_{10}$ and then restart. The counter should have a synchronous (not an asynchronous) active-high reset signal that resets the counter to '0'. You may use any logic gates/blocks learned in class and the constants '0' and '1'.

Answer